

S1 appendix for

**The Impact of Groupwork on Undergraduate State Anxiety in Science: *“It’s like willingly going into a dark forest where anything bad can happen.”***

**This supplemental material contains the following:**

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**Aspects hypothesized to alleviate undergraduate state anxiety and the literature supporting each.**

| Aspect                                                            | Supporting literature                                                                     |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Having other students support you academically                    | (Cooper et al., 2018; Downing et al., 2020).4/19/26 11:55:00 AM                           |
| Realizing that other students struggle with content too           | (Cooper et al., 2018).                                                                    |
| Being able to work with other students who you know               | (Baldock & White, 2026; Cooper et al., 2018; Downing et al., 2020; Strauss et al., 2011). |
| Having another student validate your contribution to a discussion | (Downing et al., 2020).                                                                   |
| Forming new personal connections with other students              | (Baldock & White, 2026; Strauss et al., 2011; Xethakis et al., 2024).                     |

**Aspects hypothesized to exacerbate undergraduate state anxiety and the literature supporting each.**

| Aspect                                                                | Supporting literature                                                                                            |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Being the center of attention                                         | (Cooper et al., 2018; Downing et al., 2020; England et al., 2019).                                               |
| Realizing that you may struggle with content more than other students | (Cooper et al., 2018).                                                                                           |
| Having your grade depend on other students' contributions             | (Burdett, 2003; Chang & Brickman, 2018; Chi & Kadandale, 2022).                                                  |
| Working with other students who you do not know                       | (Baldock & White, 2026; Cooper et al., 2018; Downing et al., 2020; Strauss et al., 2011; Xethakis et al., 2024). |
| Making a mistake in front of other students                           | (Downing et al., 2020; Luan et al., 2022; Xethakis et al., 2024).                                                |
| Feeling like you have talked too much                                 | (Cooper et al., 2018).                                                                                           |
| Feeling like you have talked too little                               | (Cooper et al., 2018).                                                                                           |
| A student responding rudely to your contribution                      | (Cooper et al., 2018; Downing et al., 2020; Luan et al., 2022)                                                   |
| Other students treating you differently than they would treat others  | (Baldock & White, 2026; Downing et al., 2020; Luan et al., 2022; Xethakis et al., 2024).                         |
| Being asked to initiate social interactions                           | (Xethakis et al., 2024).                                                                                         |
| Not being able to be your authentic self                              | (Van den Bosch & Taris, 2018; Van Den Bosch & Taris, 2014)                                                       |

**Learning domains hypothesized to be impacted by undergraduate state anxiety and the literature supporting each.**

| <b>Learning domains</b>                | <b>Supporting literature</b>                                        |
|----------------------------------------|---------------------------------------------------------------------|
| Remember content that you learned      | (Dowker et al., 2016; Hashempour & Mehrad, 2014; Lau et al., 2022). |
| Stay focused                           | (Baldock & White, 2026; Cooper et al., 2018).                       |
| Be motivated to engage in discussions  | (Downing et al., 2020).                                             |
| Be patient with others                 | (Rozi, 2021).                                                       |
| Communicate your thoughts              | (Cooper et al., 2018; Downing et al., 2020; Xethakis et al., 2024). |
| Be willing to contribute               | (Cooper et al., 2018; Downing et al., 2020).                        |
| Understand what others are saying      | (Xethakis et al., 2024).                                            |
| Solve problems                         | (Cooper et al., 2018).4/19/26 11:55:00 AM                           |
| Feel confident about your intelligence | (Downing et al., 2020; Luan et al., 2022).                          |

## Survey

Please think of one college science course that you are taking that involves group work. For the remainder of the survey, all questions will be about your experience in that specific class.

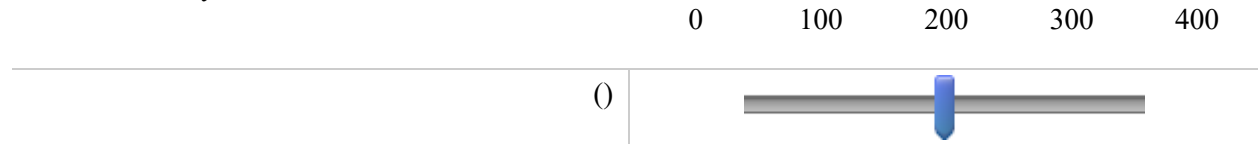
What is the primary subject of the course?

- Astronomy
- Biology
- Biochemistry
- Chemistry
- Environmental science
- Geosciences
- Physics
- Other \_\_\_\_\_

What level is this course?

- Upper level (300/400)
- Lower level (100/200)

About how many students are in this course?



What is the average number of other students (not counting yourself) who you work with during groupwork in this class?

- (1)
- (2)
- (3)
- (4)
- (5) or more

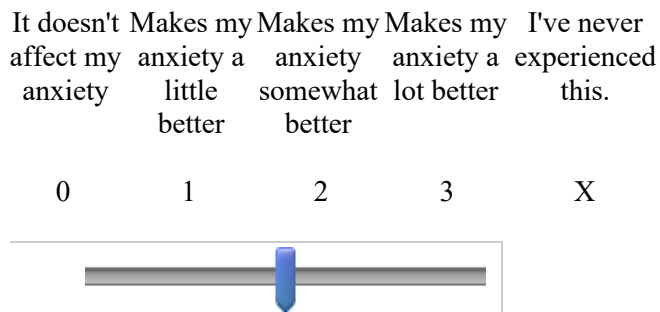
Everyone has feelings of anxiety to some extent. For some students, these feelings are short-lived and do not affect them much in class. For others, they are more severe.

Please fill out the following questions about your anxiety during the past two weeks in your college science course. Over the last 2 weeks, how often have you been bothered by the following problems in your college science course?

|                                                   | Not at all (0)        | Several days (1)      | Over half the days (2) | Nearly every day (3)  |
|---------------------------------------------------|-----------------------|-----------------------|------------------------|-----------------------|
| Feeling nervous, anxious, or on edge              | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>  | <input type="radio"/> |
| Not being able to stop or control worrying        | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>  | <input type="radio"/> |
| Worrying too much about different things          | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>  | <input type="radio"/> |
| Trouble relaxing                                  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>  | <input type="radio"/> |
| Being so restless that it's hard to sit still     | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>  | <input type="radio"/> |
| Becoming easily annoyed or irritable              | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>  | <input type="radio"/> |
| Feeling afraid as if something awful might happen | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>  | <input type="radio"/> |

The next set of questions will ask you to what extent aspects of groupwork make your feelings of anxiety **better**. When we refer to feelings of anxiety, we are talking about feeling nervous, anxious, on edge, afraid, or worried. Please answer the following questions based on your experience in the one science course you indicated at the beginning of the survey.

During groupwork, how, if at all, **does having other students support you academically** (e.g., answer your questions, discuss topics with you, help you understand content) affect your feelings of anxiety?

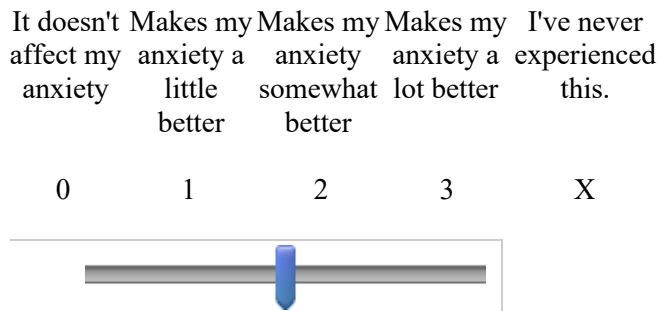


*Students who answered 2 or greater were displayed this question:*

Please explain why **having other students support you academically** makes your anxiety better.

---

During groupwork, how, if at all, **does realizing that other students struggle with content too** affect your feelings of anxiety?

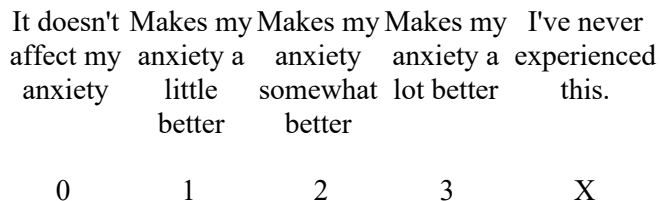


*Students who answered 2 or greater were displayed this question:*

Please explain why **realizing that other students struggle with content too** makes your anxiety better.

---

During groupwork, how, if at all, **does being able to work with other students who you know** affect your feelings of anxiety?





*Students who answered 2 or greater were displayed this question:*

Please explain why **being able to work with other students who you know** makes your anxiety better.

During groupwork, how, if at all, **does having another student validate your contribution to a discussion** (e.g., say ‘that’s a good point/question, that’s interesting, I agree with you, etc...?’) affect your feelings of anxiety?

|                                    |                                           |                                           |                                     |                                    |
|------------------------------------|-------------------------------------------|-------------------------------------------|-------------------------------------|------------------------------------|
| It doesn't<br>affect my<br>anxiety | Makes my<br>anxiety a<br>little<br>better | Makes my<br>anxiety<br>somewhat<br>better | Makes my<br>anxiety a<br>lot better | I've never<br>experienced<br>this. |
|------------------------------------|-------------------------------------------|-------------------------------------------|-------------------------------------|------------------------------------|

|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | X |
|---|---|---|---|---|



*Students who answered 2 or greater were displayed this question:*

Please explain why **having another student validate your contribution to a discussion** makes your anxiety better.

During groupwork, how, if at all, **does forming new personal connections with other students** affect your feelings of anxiety?

|                                    |                                           |                                           |                                     |                                    |
|------------------------------------|-------------------------------------------|-------------------------------------------|-------------------------------------|------------------------------------|
| It doesn't<br>affect my<br>anxiety | Makes my<br>anxiety a<br>little<br>better | Makes my<br>anxiety<br>somewhat<br>better | Makes my<br>anxiety a<br>lot better | I've never<br>experienced<br>this. |
|------------------------------------|-------------------------------------------|-------------------------------------------|-------------------------------------|------------------------------------|

|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | X |
|---|---|---|---|---|



*Students who answered 2 or greater were displayed this question:*

Please explain why you feel that **forming new personal connections with other students** makes your anxiety better.

The next set of questions will ask you to what extent aspects of groupwork make your feelings of anxiety **worse**. When we refer to feelings of anxiety, we are talking about feeling nervous, anxious, on edge,

afraid, or worried. Please answer the following questions based on your experience in the one science course you indicated at the beginning of the survey.

During groupwork, how, if at all, **does being the center of attention** affect your feelings of anxiety?

|                                    |                                          |                                          |                                    |                                    |
|------------------------------------|------------------------------------------|------------------------------------------|------------------------------------|------------------------------------|
| It doesn't<br>affect my<br>anxiety | Makes my<br>anxiety a<br>little<br>worse | Makes my<br>anxiety<br>somewhat<br>worse | Makes my<br>anxiety a<br>lot worse | I've never<br>experienced<br>this. |
|------------------------------------|------------------------------------------|------------------------------------------|------------------------------------|------------------------------------|

|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | X |
|---|---|---|---|---|



*Students who answered 2 or greater were displayed this question:*

Please explain why **being the center of attention** makes your anxiety worse.

---

During groupwork, how, if at all, **does realizing that you may struggle with content more than other students** affect your feelings of anxiety?

|                                    |                                          |                                          |                                    |                                    |
|------------------------------------|------------------------------------------|------------------------------------------|------------------------------------|------------------------------------|
| It doesn't<br>affect my<br>anxiety | Makes my<br>anxiety a<br>little<br>worse | Makes my<br>anxiety<br>somewhat<br>worse | Makes my<br>anxiety a<br>lot worse | I've never<br>experienced<br>this. |
|------------------------------------|------------------------------------------|------------------------------------------|------------------------------------|------------------------------------|

|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | X |
|---|---|---|---|---|



*Students who answered 2 or greater were displayed this question:*

Please explain why **realizing that you may struggle with content more than other students** makes your anxiety worse.

---

During groupwork, how, if at all, **does having your grade depend on other students' contributions** affect your feelings of anxiety?

|                                    |                                          |                                          |                                    |                                    |
|------------------------------------|------------------------------------------|------------------------------------------|------------------------------------|------------------------------------|
| It doesn't<br>affect my<br>anxiety | Makes my<br>anxiety a<br>little<br>worse | Makes my<br>anxiety<br>somewhat<br>worse | Makes my<br>anxiety a<br>lot worse | I've never<br>experienced<br>this. |
|------------------------------------|------------------------------------------|------------------------------------------|------------------------------------|------------------------------------|

|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | X |
|---|---|---|---|---|





*Students who answered 2 or greater were displayed this question:*

Please explain why **having your grade depend on other students' contributions** makes your anxiety worse.

---

During groupwork, how, if at all, **does working with other students who you do NOT know** affect your feelings of anxiety?

|                              |                                 |                                 |                              |                              |
|------------------------------|---------------------------------|---------------------------------|------------------------------|------------------------------|
| It doesn't affect my anxiety | Makes my anxiety a little worse | Makes my anxiety somewhat worse | Makes my anxiety a lot worse | I've never experienced this. |
|------------------------------|---------------------------------|---------------------------------|------------------------------|------------------------------|

|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | X |
|---|---|---|---|---|



*Students who answered 2 or greater were displayed this question:*

Please explain why **working with other students who you do NOT know** makes your anxiety worse.

---

During groupwork, how, if at all, **does making a mistake in front of other students** affect your feelings of anxiety?

|                              |                                 |                                 |                              |                              |
|------------------------------|---------------------------------|---------------------------------|------------------------------|------------------------------|
| It doesn't affect my anxiety | Makes my anxiety a little worse | Makes my anxiety somewhat worse | Makes my anxiety a lot worse | I've never experienced this. |
|------------------------------|---------------------------------|---------------------------------|------------------------------|------------------------------|

|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | X |
|---|---|---|---|---|

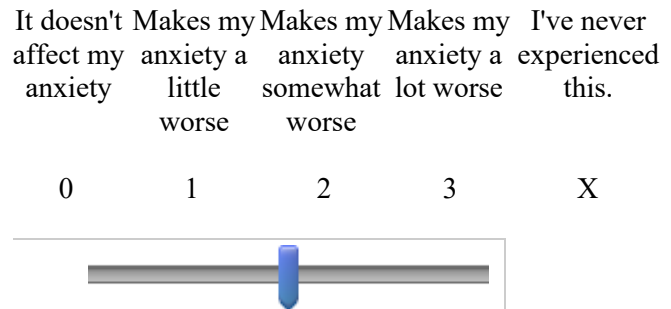


*Students who answered 2 or greater were displayed this question:*

Please explain why **making a mistake in front of other students** makes your anxiety worse.

---

During groupwork, how, if at all, **does feeling like you have talked too much** affect your feelings of anxiety?

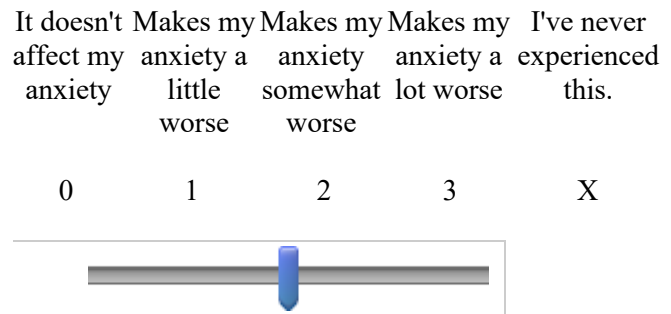


*Students who answered 2 or greater were displayed this question:*

Please explain why **feeling like you have talked too much** makes your anxiety worse.

---

During groupwork, how, if at all, **does feeling like you have talked too LITTLE** affect your feelings of anxiety?

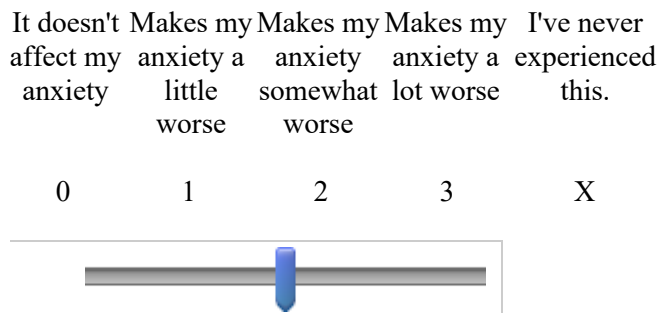


*Students who answered 2 or greater were displayed this question:*

Please explain why **feeling like you have talked too LITTLE** makes your anxiety worse.

---

During groupwork, how, if at all, **does a student responding rudely to your contribution** affect your feelings of anxiety?

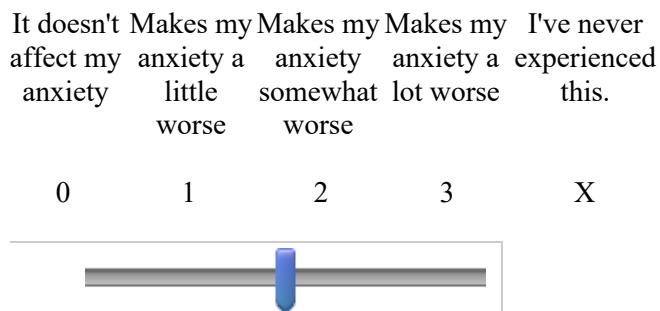


*Students who answered 2 or greater were displayed this question:*

Please explain why **a student responding rudely to your contribution** makes your anxiety worse.

---

During groupwork, how, if at all, **does other students treating you differently than they would treat others** (potentially based on identity, abilities, appearance, personality, etc...) affect your feelings of anxiety?

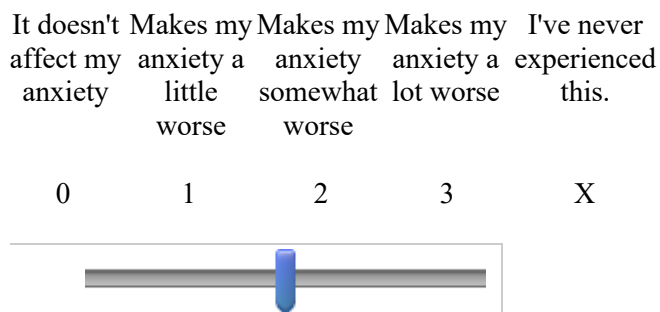


*Students who answered 2 or greater were displayed this question:*

Please explain why **other students treating you differently than they would treat others** makes your anxiety worse.

---

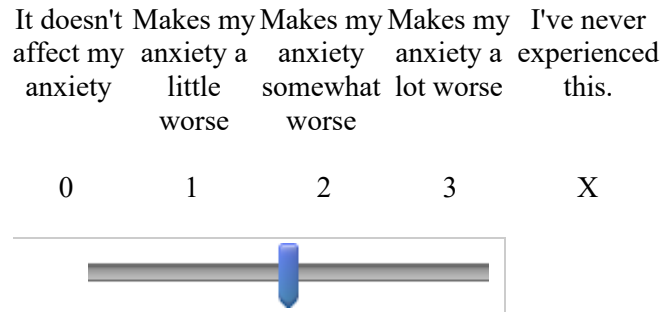
During groupwork, how, if at all, **does being asked to initiate social interactions (find a group)** affect your feelings of anxiety?



*Students who answered 2 or greater were displayed this question:*

Please explain why **being asked to initiate social interactions (find a group)** makes your anxiety worse.

During groupwork, how, if at all, **does not being able to be your authentic self** affect your feelings of anxiety?



*Students who answered 2 or greater were displayed this question:*

Please explain why **not being able to be your authentic self** makes your anxiety worse.

Please indicate how your **feelings of anxiety** impact **your ability to do the following during groupwork** in your college science course.

*For example, for the first row you should indicate whether your anxiety makes it harder or easier for you to remember content that you learned during groupwork.*

|                                                         | Makes it a lot harder | Makes it a little harder | Doesn't change my ability | Makes it a little easier | Makes it a lot easier |
|---------------------------------------------------------|-----------------------|--------------------------|---------------------------|--------------------------|-----------------------|
| Remember content that you learned                       | <input type="radio"/> | <input type="radio"/>    | <input type="radio"/>     | <input type="radio"/>    | <input type="radio"/> |
| Stay focused during groupwork                           | <input type="radio"/> | <input type="radio"/>    | <input type="radio"/>     | <input type="radio"/>    | <input type="radio"/> |
| Be motivated to engage in discussions during groupwork  | <input type="radio"/> | <input type="radio"/>    | <input type="radio"/>     | <input type="radio"/>    | <input type="radio"/> |
| Be patient with others during groupwork                 | <input type="radio"/> | <input type="radio"/>    | <input type="radio"/>     | <input type="radio"/>    | <input type="radio"/> |
| Communicate your thoughts during groupwork              | <input type="radio"/> | <input type="radio"/>    | <input type="radio"/>     | <input type="radio"/>    | <input type="radio"/> |
| Your willingness to contribute during groupwork         | <input type="radio"/> | <input type="radio"/>    | <input type="radio"/>     | <input type="radio"/>    | <input type="radio"/> |
| Understand what others are saying during groupwork      | <input type="radio"/> | <input type="radio"/>    | <input type="radio"/>     | <input type="radio"/>    | <input type="radio"/> |
| Solving problems during groupwork                       | <input type="radio"/> | <input type="radio"/>    | <input type="radio"/>     | <input type="radio"/>    | <input type="radio"/> |
| Feel confident about your intelligence during groupwork | <input type="radio"/> | <input type="radio"/>    | <input type="radio"/>     | <input type="radio"/>    | <input type="radio"/> |

In your opinion, what can primary instructors do to help decrease students' feelings of anxiety associated with groupwork?

---

Our lab group focuses on creating a more inclusive scientific environment for all students. As such, it is helpful for us to know more about you. You may skip any question you do not want to answer.

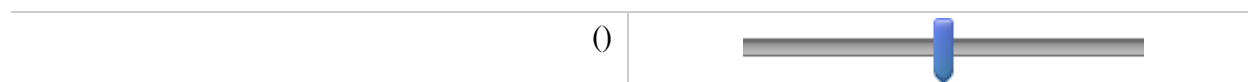
What field is your major most closely associated with? If you have more than one major, please choose all that apply.

- Biology
- Physics
- Geoscience
- Chemistry or biochemistry
- Engineering or technology
- Nursing, public health, or medical studies
- Business
- Literature, art, or design
- Another field not listed here \_\_\_\_\_

What is your GPA (on a 4.0 scale)? If this is your first semester, please select "no GPA".

no GPA

0 1 2 3 4



I most closely identify as

- Woman
- Man
- Gender-queer or non-binary
- An identity not listed, please specify \_\_\_\_\_
- Decline to state

I most closely identify as

- American Indian or Alaska Native
  - Asian or Asian American
  - Black or African American
  - Middle Eastern or Middle Eastern American
  - Hispanic, Latina/e, or of Spanish origin
  - Pacific Islander
  - White
  - An identity not listed including multiracial, please describe
- 
- Decline to state

What is your parent's highest completed level of education? If you have more than one parent with differing levels of education, choose the higher of the two.

- Did not complete high school
  - High school diploma or GED
  - Some college but no degree
  - Associate degree (for example: AA, AS)
  - Bachelor's degree (for example: BA, AB, BS)
  - Master's degree (for example: MA, MS, MEng, MEd, MSW, MBA)
  - Higher than a Master's degree (for example: PhD, MD, JD)
  - Other (please describe)
- \_\_\_\_\_
- Decline to state

Do you currently feel financially stable (e.g. have enough money to pay for necessities such as food and rent).

- Yes
- Yes, but only sometimes
- No
- Decline to state

Do you identify as a member of the LGBTQIA+\* community?

*\*lesbian, gay, bisexual, transgender, queer/questioning, intersex, asexual, and more*

- Yes
- No
- Decline to state

I most closely identify with

- Currently or having previously struggled with anxiety or an anxiety disorder
- I have never struggled with an anxiety disorder
- Decline to state

## RQ1a: output for final cross-validation models

Raw output of final models from cross-validation selection models for RQ1a outcomes; aligns with Table 1 of the main text.

A cross-validated selection procedure was run on each unique outcome, starting with identical predictors and model settings.

Initial foundational model:

```
initial.model.template <- caret::train( [outcome] ~ gender +  
lgbtq + race + firstgen + gpa.mean.centered + finance +  
class.size.scale + class.discipline, data = data.RQ1.analysis,  
method = "lmStepAIC", direction = "backward", trControl = ctrl)
```

### Being able to work with other students who you know

Call:

```
lm(formula = .outcome ~ lgbtqLGBTQ + racePEER + gpa.mean.centered +  
class.disciplineChemistry, data = dat)
```

Residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -2.0111 | -0.5300 | 0.2299 | 0.5273 | 1.5295 |

Coefficients:

|                           | Estimate | Std. Error | t value | Pr(> t )    |
|---------------------------|----------|------------|---------|-------------|
| (Intercept)               | 1.6816   | 0.1015     | 16.569  | < 2e-16 *** |
| lgbtqLGBTQ                | 0.4049   | 0.1679     | 2.412   | 0.01717 *   |
| racePEER                  | 0.2940   | 0.1455     | 2.021   | 0.04524 *   |
| gpa.mean.centered         | 0.5922   | 0.1695     | 3.494   | 0.00064 *** |
| class.disciplineChemistry | 0.3615   | 0.1846     | 1.958   | 0.05227 .   |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8457 on 139 degrees of freedom

Multiple R-squared: 0.1626, Adjusted R-squared: 0.1385

F-statistic: 6.746 on 4 and 139 DF, p-value: 5.429e-05



**Having another student validate your contribution to a discussion**

Call:

```
lm(formula = .outcome ~ 1, data = dat)
```

Residuals:

| Min      | 1Q       | Median  | 3Q      | Max     |
|----------|----------|---------|---------|---------|
| -1.95833 | -0.75833 | 0.04167 | 1.04167 | 1.04167 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | 1.95833  | 0.07512    | 26.07   | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9014 on 143 degrees of freedom

### Realizing that other students struggle with the content too

Call:

```
lm(formula = .outcome ~ genderWoman.NB + lgbtqLGBTQ + raceAsian +  
    gpa.mean.centered, data = dat)
```

Residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -2.3578 | -0.5427 | 0.1863 | 0.7752 | 1.3582 |

Coefficients:

|                   | Estimate | Std. Error | t value | Pr(> t ) |     |
|-------------------|----------|------------|---------|----------|-----|
| (Intercept)       | 1.6973   | 0.1540     | 11.022  | <2e-16   | *** |
| genderWoman.NB    | 0.3028   | 0.1724     | 1.757   | 0.0811   | .   |
| lgbtqLGBTQ        | 0.3340   | 0.1840     | 1.815   | 0.0716   | .   |
| raceAsian         | -0.3619  | 0.1998     | -1.812  | 0.0722   | .   |
| gpa.mean.centered | 0.3958   | 0.1870     | 2.117   | 0.0360   | *   |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9352 on 139 degrees of freedom

Multiple R-squared: 0.09906, Adjusted R-squared: 0.07313

F-statistic: 3.821 on 4 and 139 DF, p-value: 0.0056

### Forming new personal connections with other students

Call:

```
lm(formula = .outcome ~ genderWoman.NB + racePEER + gpa.mean.centered,  
    data = dat)
```

Residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -1.9313 | -0.5659 | 0.1432 | 0.5437 | 1.6347 |

Coefficients:

|                   | Estimate | Std. Error | t value | Pr(> t )    |
|-------------------|----------|------------|---------|-------------|
| (Intercept)       | 1.4200   | 0.1397     | 10.161  | < 2e-16 *** |
| genderWoman.NB    | 0.2681   | 0.1568     | 1.710   | 0.08942 .   |
| racePEER          | 0.4149   | 0.1461     | 2.840   | 0.00518 **  |
| gpa.mean.centered | 0.3901   | 0.1708     | 2.284   | 0.02387 *   |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8489 on 140 degrees of freedom

Multiple R-squared: 0.1056, Adjusted R-squared: 0.08648

F-statistic: 5.512 on 3 and 140 DF, p-value: 0.001315

### Having other students support you academically

Call:

```
lm(formula = .outcome ~ gpa.mean.centered, data = dat)
```

Residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -1.8041 | -0.6577 | 0.1958 | 0.5389 | 1.4246 |

Coefficients:

|                   | Estimate | Std. Error | t value | Pr(> t )   |
|-------------------|----------|------------|---------|------------|
| (Intercept)       | 1.67267  | 0.07397    | 22.614  | <2e-16 *** |
| gpa.mean.centered | 0.28589  | 0.17679    | 1.617   | 0.108      |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8871 on 142 degrees of freedom

Multiple R-squared: 0.01808, Adjusted R-squared: 0.01117

F-statistic: 2.615 on 1 and 142 DF, p-value: 0.1081

## RQ2a: output for final cross-validation models

Raw output of final models from cross-validation selection models for RQ2a outcomes; aligns with Table 3 of the main text.

A cross-validated selection procedure was run on each unique outcome, starting with identical predictors and model settings.

Initial foundational model:

```
initial.model.template <- caret::train( [outcome] ~ gender +  
lgbtq + race + firstgen + gpa.mean.centered + finance +  
class.size.scale + class.discipline, data = data.RQ2.analysis,  
method = "lmStepAIC", direction = "backward", trControl = ctrl)
```

### Having your grade depend on other students' contributions

Call:

```
lm(formula = .outcome ~ firstgenYes + class.size.scale, data =  
data.RQ2.analysis)
```

Residuals:

| Min      | 1Q       | Median  | 3Q      | Max     |
|----------|----------|---------|---------|---------|
| -2.04982 | -0.62009 | 0.01733 | 0.79930 | 1.65144 |

Coefficients:

|                  | Estimate | Std. Error | t value | Pr(> t )   |
|------------------|----------|------------|---------|------------|
| (Intercept)      | 1.97081  | 0.11123    | 17.718  | <2e-16 *** |
| firstgenYes      | -0.43131 | 0.17811    | -2.422  | 0.0173 *   |
| class.size.scale | -0.13523 | 0.08805    | -1.536  | 0.1278     |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8738 on 99 degrees of freedom

Multiple R-squared: 0.0749, Adjusted R-squared: 0.05621

F-statistic: 4.008 on 2 and 99 DF, p-value: 0.0212

### **A student responding rudely to your contribution**

Call:

```
lm(formula = .outcome ~ genderWoman.NB + lgbtqLGBTQ + racePEER +  
    firstgenYes + financeUnstable + class.disciplinePhysics +  
    class.disciplineChemistry, data = data.RQ2.analysis)
```

Residuals:

| Min      | 1Q       | Median  | 3Q      | Max     |
|----------|----------|---------|---------|---------|
| -1.76005 | -0.64571 | 0.00942 | 0.63876 | 1.96790 |

Coefficients:

|                           | Estimate | Std. Error | t value | Pr(> t ) |     |
|---------------------------|----------|------------|---------|----------|-----|
| (Intercept)               | 0.6958   | 0.2327     | 2.990   | 0.003557 | **  |
| genderWoman.NB            | 0.6268   | 0.1943     | 3.226   | 0.001727 | **  |
| lgbtqLGBTQ                | 0.5906   | 0.2077     | 2.844   | 0.005467 | **  |
| racePEER                  | 0.2762   | 0.1868     | 1.478   | 0.142734 |     |
| firstgenYes               | -0.7290  | 0.1998     | -3.648  | 0.000434 | *** |
| financeUnstable           | 0.5197   | 0.1898     | 2.738   | 0.007387 | **  |
| class.disciplinePhysics   | 0.3363   | 0.2107     | 1.596   | 0.113803 |     |
| class.disciplineChemistry | 0.3469   | 0.2371     | 1.463   | 0.146787 |     |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8437 on 94 degrees of freedom

Multiple R-squared: 0.2737, Adjusted R-squared: 0.2196

F-statistic: 5.06 on 7 and 94 DF, p-value: 6.676e-05

**Realizing that you may struggle with content more than other students**

Call:

```
lm(formula = .outcome ~ genderWoman.NB + class.size.scale, data =  
data.RQ2.analysis)
```

Residuals:

| Min      | 1Q       | Median  | 3Q      | Max     |
|----------|----------|---------|---------|---------|
| -2.08391 | -0.81533 | 0.04815 | 0.89717 | 2.04300 |

Coefficients:

|                  | Estimate | Std. Error | t value | Pr(> t ) |     |
|------------------|----------|------------|---------|----------|-----|
| (Intercept)      | 1.03629  | 0.17383    | 5.961   | 3.86e-08 | *** |
| genderWoman.NB   | 0.76742  | 0.20569    | 3.731   | 0.000318 | *** |
| class.size.scale | -0.19868 | 0.09624    | -2.064  | 0.041598 | *   |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9525 on 99 degrees of freedom

Multiple R-squared: 0.1471, Adjusted R-squared: 0.1298

F-statistic: 8.535 on 2 and 99 DF, p-value: 0.0003805

# Other students treating you differently than they would treat others

Call:

```
lm(formula = .outcome ~ genderWoman.NB + lgbtqLGBTQ + raceAsian +  
    racePEER + financeUnstable + class.size.scale, data =  
data.RQ2.analysis)
```

Residuals:

| Min      | 1Q       | Median  | 3Q      | Max     |
|----------|----------|---------|---------|---------|
| -1.82301 | -0.48859 | 0.03966 | 0.46389 | 2.21346 |

Coefficients:

|                  | Estimate | Std. Error | t value | Pr(> t ) |     |
|------------------|----------|------------|---------|----------|-----|
| (Intercept)      | 0.48711  | 0.18966    | 2.568   | 0.01178  | *   |
| genderWoman.NB   | 0.64374  | 0.15556    | 4.138   | 7.57e-05 | *** |
| lgbtqLGBTQ       | 0.47879  | 0.17449    | 2.744   | 0.00726  | **  |
| raceAsian        | 0.41705  | 0.19418    | 2.148   | 0.03428  | *   |
| racePEER         | 0.26129  | 0.16431    | 1.590   | 0.11510  |     |
| financeUnstable  | 0.37823  | 0.15447    | 2.449   | 0.01618  | *   |
| class.size.scale | -0.21804 | 0.07436    | -2.932  | 0.00422  | **  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7194 on 95 degrees of freedom

Multiple R-squared: 0.3361, Adjusted R-squared: 0.2942

F-statistic: 8.015 on 6 and 95 DF, p-value: 5.22e-07



### **Making a mistake in front of other students**

Call:

```
lm(formula = .outcome ~ 1, data = data.RQ2.analysis)
```

Residuals:

| Min      | 1Q       | Median  | 3Q      | Max     |
|----------|----------|---------|---------|---------|
| -1.48529 | -0.58529 | 0.01471 | 0.51471 | 1.51471 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | 1.48529  | 0.09011    | 16.48   | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9101 on 101 degrees of freedom

### Being the center of attention

Call:

```
lm(formula = .outcome ~ genderWoman.NB, data = data.RQ2.analysis)
```

Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -1.4718 | -0.5718 | -0.0714 | 0.6034 | 1.8290 |

Coefficients:

|                | Estimate | Std. Error | t value | Pr(> t )     |
|----------------|----------|------------|---------|--------------|
| (Intercept)    | 1.1710   | 0.1668     | 7.022   | 2.69e-10 *** |
| genderWoman.NB | 0.3009   | 0.1999     | 1.505   | 0.135        |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9285 on 100 degrees of freedom

Multiple R-squared: 0.02216, Adjusted R-squared: 0.01238

F-statistic: 2.266 on 1 and 100 DF, p-value: 0.1354

### Being asked to initiate social interactions

Call:

```
lm(formula = .outcome ~ genderWoman.NB + financeUnstable, data =  
data.RQ2.analysis)
```

Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -1.6685 | -0.7087 | -0.0953 | 0.6750 | 1.6663 |

Coefficients:

|                 | Estimate | Std. Error | t value | Pr(> t )     |
|-----------------|----------|------------|---------|--------------|
| (Intercept)     | 0.7872   | 0.2051     | 3.838   | 0.000219 *** |
| genderWoman.NB  | 0.5465   | 0.1958     | 2.791   | 0.006314 **  |
| financeUnstable | 0.3349   | 0.1925     | 1.739   | 0.085112 .   |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9088 on 99 degrees of freedom

Multiple R-squared: 0.1021, Adjusted R-squared: 0.08401

F-statistic: 5.632 on 2 and 99 DF, p-value: 0.004827

### Working with other students who you do not know

Call:

```
lm(formula = .outcome ~ raceAsian + racePEER + firstgenYes, data =  
data.RQ2.analysis)
```

Residuals:

| Min      | 1Q       | Median  | 3Q      | Max     |
|----------|----------|---------|---------|---------|
| -1.46632 | -0.50717 | 0.03148 | 0.54028 | 1.87921 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 1.1208   | 0.1383     | 8.104   | 1.52e-12 *** |
| raceAsian   | 0.3389   | 0.2167     | 1.564   | 0.1211       |
| racePEER    | 0.3455   | 0.1997     | 1.730   | 0.0867 .     |
| firstgenYes | -0.2956  | 0.1821     | -1.623  | 0.1078       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8147 on 98 degrees of freedom

Multiple R-squared: 0.04984, Adjusted R-squared: 0.02075

F-statistic: 1.713 on 3 and 98 DF, p-value: 0.1692

### Feeling like you have talked too much

Call:

```
lm(formula = .outcome ~ lgbtqLGBTQ + raceAsian + gpa.mean.centered,  
    data = data.RQ2.analysis)
```

Residuals:

| Min      | 1Q       | Median  | 3Q      | Max     |
|----------|----------|---------|---------|---------|
| -1.55463 | -0.76551 | 0.03375 | 0.54827 | 2.06243 |

Coefficients:

|                   | Estimate | Std. Error | t value | Pr(> t )     |
|-------------------|----------|------------|---------|--------------|
| (Intercept)       | 0.8974   | 0.1059     | 8.473   | 2.45e-13 *** |
| lgbtqLGBTQ        | 0.4737   | 0.2044     | 2.317   | 0.0226 *     |
| raceAsian         | 0.3510   | 0.2031     | 1.728   | 0.0872 .     |
| gpa.mean.centered | -0.2868  | 0.2049     | -1.399  | 0.1649       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8491 on 98 degrees of freedom

Multiple R-squared: 0.09126, Adjusted R-squared: 0.06344

F-statistic: 3.281 on 3 and 98 DF, p-value: 0.02414

### Not being able to be your authentic self

Call:

```
lm(formula = .outcome ~ class.size.scale, data = data.RQ2.analysis)
```

Residuals:

| Min      | 1Q       | Median   | 3Q      | Max     |
|----------|----------|----------|---------|---------|
| -1.25687 | -0.64660 | -0.04993 | 0.54333 | 2.08075 |

Coefficients:

|                  | Estimate | Std. Error | t value | Pr(> t )   |
|------------------|----------|------------|---------|------------|
| (Intercept)      | 1.03319  | 0.08387    | 12.319  | <2e-16 *** |
| class.size.scale | -0.15535 | 0.08356    | -1.859  | 0.0659 .   |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8295 on 100 degrees of freedom

Multiple R-squared: 0.03341, Adjusted R-squared: 0.02375

F-statistic: 3.457 on 1 and 100 DF, p-value: 0.06593

### Feeling like you have talked too little

Call:

```
lm(formula = .outcome ~ genderWoman.NB + financeUnstable, data =  
data.RQ2.analysis)
```

Residuals:

| Min      | 1Q       | Median   | 3Q      | Max     |
|----------|----------|----------|---------|---------|
| -1.16303 | -0.55242 | -0.00068 | 0.54181 | 2.17675 |

Coefficients:

|                 | Estimate | Std. Error | t value | Pr(> t )   |
|-----------------|----------|------------|---------|------------|
| (Intercept)     | 0.4808   | 0.1643     | 2.926   | 0.00426 ** |
| genderWoman.NB  | 0.3425   | 0.1569     | 2.183   | 0.03141 *  |
| financeUnstable | 0.3398   | 0.1543     | 2.203   | 0.02993 *  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7281 on 99 degrees of freedom

Multiple R-squared: 0.09226, Adjusted R-squared: 0.07392

F-statistic: 5.031 on 2 and 99 DF, p-value: 0.0083

### RQ3a: output for final stepwise selection models

Raw output of final models for stepwise ordinal regressions for RQ3 outcomes; aligns with Table 5 in the main text.

```
ordmod.outcome <- ordinal::clm(outcome.fact ~ gender + lgbtq +  
  race + firstgen + gpa.mean.centered + finance + class.size.scale  
  + class.discipline + gad.avg, data = data.RQ3.analysis, Hess =  
  TRUE)
```

```
step.ord.outcome <- step(ordmod.outcome, direction = "backward")
```

### Communicate your thoughts

```
formula: AonAL_communicate.fact ~ gender + firstgen + class.size.scale  
+ class.discipline  
data:    data.RQ3.analysis
```

```
link threshold nobs logLik AIC      niter max.grad cond.H  
logit flexible 178 -249.21 516.43 5(0) 8.55e-08 3.5e+01
```

#### Coefficients:

|                           | Estimate | Std. Error | z value | Pr(> z ) |
|---------------------------|----------|------------|---------|----------|
| genderWoman.NB            | -0.4227  | 0.2902     | -1.457  | 0.1452   |
| firstgenYes               | 0.5396   | 0.2949     | 1.830   | 0.0673 . |
| class.size.scale          | 0.4171   | 0.1622     | 2.571   | 0.0101 * |
| class.disciplinePhysics   | 0.8763   | 0.3886     | 2.255   | 0.0241 * |
| class.disciplineChemistry | 0.4175   | 0.4436     | 0.941   | 0.3467   |

---

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#### Threshold coefficients:

|       | Estimate | Std. Error | z value |
|-------|----------|------------|---------|
| -2 -1 | -2.08642 | 0.33160    | -6.292  |
| -1 0  | 0.07327  | 0.27285    | 0.269   |
| 0 1   | 1.42321  | 0.29559    | 4.815   |
| 1 2   | 2.82796  | 0.38601    | 7.326   |



### Feel confident about your intelligence

formula:

```
AonAL_confident.fact ~ gender + firstgen + gpa.mean.centered +  
class.size.scale + class.discipline
```

data: data.RQ3.analysis

```
link threshold nobs logLik AIC niter max.grad cond.H  
logit flexible 178 -262.04 544.08 5(0) 6.29e-13 3.6e+01
```

Coefficients:

|                           | Estimate | Std. Error | z value | Pr(> z ) |    |
|---------------------------|----------|------------|---------|----------|----|
| genderWoman.NB            | -0.8287  | 0.2908     | -2.850  | 0.00438  | ** |
| firstgenYes               | 0.7673   | 0.2972     | 2.582   | 0.00983  | ** |
| gpa.mean.centered         | 0.5810   | 0.3293     | 1.764   | 0.07769  | .  |
| class.size.scale          | 0.3258   | 0.1609     | 2.025   | 0.04287  | *  |
| class.disciplinePhysics   | 0.7871   | 0.3929     | 2.003   | 0.04515  | *  |
| class.disciplineChemistry | 0.1734   | 0.4412     | 0.393   | 0.69440  |    |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:

|       | Estimate | Std. Error | z value |
|-------|----------|------------|---------|
| -2 -1 | -1.6103  | 0.2993     | -5.381  |
| -1 0  | -0.1184  | 0.2662     | -0.445  |
| 0 1   | 1.2096   | 0.2799     | 4.321   |
| 1 2   | 2.0421   | 0.3218     | 6.346   |

**Be willing to contribute**

formula: AonAL\_contribute.fact ~ firstgen

data: data.RQ3.analysis

| link  | threshold | nobs | logLik  | AIC    | niter | max.grad | cond.H  |
|-------|-----------|------|---------|--------|-------|----------|---------|
| logit | flexible  | 178  | -265.99 | 541.99 | 5(0)  | 2.18e-11 | 1.5e+01 |

## Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z ) |
|-------------|----------|------------|---------|----------|
| firstgenYes | 0.5397   | 0.2893     | 1.866   | 0.0621   |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

## Threshold coefficients:

|       | Estimate | Std. Error | z value |
|-------|----------|------------|---------|
| -2 -1 | -2.1593  | 0.2740     | -7.882  |
| -1 0  | -0.5373  | 0.1815     | -2.961  |
| 0 1   | 0.9647   | 0.1920     | 5.025   |
| 1 2   | 2.1118   | 0.2519     | 8.384   |

**Be motivated to engage in discussions**

formula: AonAL\_engage.fact ~ class.size.scale + gad.avg

data: data.RQ3.analysis

```
link threshold nobs logLik AIC niter max.grad cond.H
logit flexible 178 -263.67 539.34 5(0) 2.12e-11 1.8e+01
```

## Coefficients:

|                  | Estimate | Std. Error | z value | Pr(> z ) |
|------------------|----------|------------|---------|----------|
| class.size.scale | 0.2796   | 0.1418     | 1.972   | 0.0487 * |
| gad.avg          | -0.2886  | 0.1791     | -1.612  | 0.1070   |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

## Threshold coefficients:

|       | Estimate | Std. Error | z value |
|-------|----------|------------|---------|
| -2 -1 | -1.9808  | 0.2614     | -7.578  |
| -1 0  | -0.2346  | 0.2043     | -1.148  |
| 0 1   | 0.9723   | 0.2233     | 4.354   |
| 1 2   | 2.1622   | 0.3017     | 7.166   |

### Stay focused

```
formula: AonAL_focus.fact ~ gender + race + firstgen +  
class.size.scale + class.discipline + gad.avg  
data:    data.RQ3.analysis
```

```
link threshold nobs logLik AIC      niter max.grad cond.H  
logit flexible 178 -242.05 508.10 5(0) 5.54e-07 7.1e+01
```

#### Coefficients:

|                           | Estimate | Std. Error | z value | Pr(> z ) |     |
|---------------------------|----------|------------|---------|----------|-----|
| genderWoman.NB            | -0.7899  | 0.3008     | -2.625  | 0.008652 | **  |
| raceAsian                 | 1.0633   | 0.3797     | 2.800   | 0.005111 | **  |
| racePEER                  | 0.8758   | 0.3352     | 2.613   | 0.008988 | **  |
| firstgenYes               | 0.4877   | 0.3116     | 1.565   | 0.117515 |     |
| class.size.scale          | 0.4773   | 0.1741     | 2.742   | 0.006108 | **  |
| class.disciplinePhysics   | 0.7691   | 0.4048     | 1.900   | 0.057411 | .   |
| class.disciplineChemistry | 1.6933   | 0.4534     | 3.735   | 0.000188 | *** |
| gad.avg                   | -0.2750  | 0.1819     | -1.512  | 0.130613 |     |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

#### Threshold coefficients:

|       | Estimate | Std. Error | z value |
|-------|----------|------------|---------|
| -2 -1 | -2.1799  | 0.3938     | -5.535  |
| -1 0  | 0.1067   | 0.3388     | 0.315   |
| 0 1   | 1.6424   | 0.3608     | 4.553   |
| 1 2   | 2.9402   | 0.4268     | 6.889   |

**Be patient with others**

```
formula: AonAL_patient.fact ~ race + firstgen + gpa.mean.centered +  
gad.avg
```

```
data: data.RQ3.analysis
```

```
link threshold nobs logLik AIC niter max.grad cond.H  
logit flexible 178 -245.20 508.41 5(0) 1.03e-07 3.6e+01
```

## Coefficients:

|                   | Estimate | Std. Error | z value | Pr(> z ) |
|-------------------|----------|------------|---------|----------|
| raceAsian         | 0.7922   | 0.3775     | 2.098   | 0.0359 * |
| racePEER          | -0.3109  | 0.3203     | -0.971  | 0.3317   |
| firstgenYes       | 0.6528   | 0.3081     | 2.119   | 0.0341 * |
| gpa.mean.centered | -0.6471  | 0.3395     | -1.906  | 0.0567 . |
| gad.avg           | -0.3274  | 0.1738     | -1.884  | 0.0595 . |

---

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## Threshold coefficients:

|       | Estimate | Std. Error | z value |
|-------|----------|------------|---------|
| -2 -1 | -2.4504  | 0.3457     | -7.088  |
| -1 0  | -0.5672  | 0.2662     | -2.131  |
| 0 1   | 1.3181   | 0.2857     | 4.614   |
| 1 2   | 2.4767   | 0.3529     | 7.018   |

**Remember content that you learned**

```
formula: AonAL_remember.fact ~ race + gad.avg
```

```
data: data.RQ3.analysis
```

```
link threshold nobs logLik AIC niter max.grad cond.H
logit flexible 178 -258.81 531.61 5(0) 1.87e-10 3.3e+01
```

**Coefficients:**

|           | Estimate | Std. Error | z value | Pr(> z ) |   |
|-----------|----------|------------|---------|----------|---|
| raceAsian | 0.6567   | 0.3684     | 1.783   | 0.0747   | . |
| racePEER  | -0.1675  | 0.3037     | -0.552  | 0.5813   |   |
| gad.avg   | -0.3673  | 0.1827     | -2.011  | 0.0443   | * |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**Threshold coefficients:**

|       | Estimate | Std. Error | z value |
|-------|----------|------------|---------|
| -2 -1 | -2.0899  | 0.3144     | -6.647  |
| -1 0  | -0.2725  | 0.2532     | -1.076  |
| 0 1   | 1.0631   | 0.2707     | 3.927   |
| 1 2   | 2.3800   | 0.3561     | 6.683   |

### Solve problems

```
formula: AonAL_solve.fact ~ gender + firstgen + class.size.scale  
data:    data.RQ3.analysis
```

```
link threshold nobs logLik AIC      niter max.grad cond.H  
logit flexible 178 -249.09 512.19 6(0) 1.23e-13 2.9e+01
```

#### Coefficients:

|                  | Estimate | Std. Error | z value | Pr(> z ) |    |
|------------------|----------|------------|---------|----------|----|
| genderWoman.NB   | -0.7694  | 0.2902     | -2.652  | 0.00801  | ** |
| firstgenYes      | 0.6001   | 0.2972     | 2.020   | 0.04343  | *  |
| class.size.scale | 0.2234   | 0.1461     | 1.529   | 0.12631  |    |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

#### Threshold coefficients:

|       | Estimate | Std. Error | z value |
|-------|----------|------------|---------|
| -2 -1 | -3.2551  | 0.3973     | -8.193  |
| -1 0  | -1.0130  | 0.2618     | -3.869  |
| 0 1   | 0.6989   | 0.2526     | 2.767   |
| 1 2   | 1.5218   | 0.2840     | 5.359   |

### Understand what others are saying

```
formula: AonAL_understand.fact ~ gender + race + firstgen +  
class.size.scale  
data:    data.RQ3.analysis
```

```
link threshold nobs logLik AIC      niter max.grad cond.H  
logit flexible 178 -244.89 507.78 6(0) 5.05e-11 3.3e+01
```

#### Coefficients:

|                  | Estimate | Std. Error | z value | Pr(> z ) |
|------------------|----------|------------|---------|----------|
| genderWoman.NB   | -0.6136  | 0.2881     | -2.130  | 0.0332 * |
| raceAsian        | 0.7287   | 0.3740     | 1.949   | 0.0513 . |
| racePEER         | 0.5187   | 0.3269     | 1.587   | 0.1126   |
| firstgenYes      | 0.4350   | 0.3076     | 1.414   | 0.1574   |
| class.size.scale | 0.2341   | 0.1487     | 1.574   | 0.1154   |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

#### Threshold coefficients:

|       | Estimate | Std. Error | z value |
|-------|----------|------------|---------|
| -2 -1 | -3.0846  | 0.4367     | -7.063  |
| -1 0  | -0.5866  | 0.2895     | -2.026  |
| 0 1   | 1.0517   | 0.2982     | 3.527   |
| 1 2   | 2.3377   | 0.3566     | 6.556   |



**Justification for variable grouping**

We combined students who identified as nonbinary (NB) with women into one category to avoid excluding the small number of nonbinary individual; compared to men, both women and NB individuals are underserved in science (Cooper et al., 2020). Similarly, we grouped American Native or Pacific Islander, Black, and Multiracial students with Hispanic students in the persons excluded because of their ethnicity or race (PEER) category (Asai, 2020), and categorized Asian and white students separately.

Demographics table

| Demographic                                   |                                     | % (n)<br>(N= 272) |
|-----------------------------------------------|-------------------------------------|-------------------|
| Gender                                        | Man                                 | 33% (89)          |
|                                               | Woman                               | 63% (172)         |
|                                               | Non-binary or gender queer          | 3% (8)            |
|                                               | <i>Decline to state</i>             | 1% (4)            |
| Race/ethnicity                                | American Native or Pacific Islander | 2% (5)            |
|                                               | Asian                               | 19% (53)          |
|                                               | Black                               | 1% (3)            |
|                                               | Hispanic                            | 26% (71)          |
|                                               | Multiracial                         | 4% (10)           |
|                                               | White                               | 40% (110)         |
|                                               | <i>Decline to state</i>             | 5% (14)           |
| LGBTQ+                                        | No                                  | 75% (203)         |
|                                               | Yes                                 | 21% (56)          |
|                                               | <i>Decline to state</i>             | 5% (13)           |
| College Generation Status                     | Continuing                          | 61% (165)         |
|                                               | First-generation                    | 35% (96)          |
|                                               | <i>Decline to state</i>             | 4% (11)           |
| Financial Stable                              | Yes                                 | 43% (116)         |
|                                               | No                                  | 52% (141)         |
|                                               | <i>Decline to state</i>             | 5% (15)           |
| GPA                                           | Mean                                | 3.54              |
|                                               | "+- std"                            | 0.45              |
| Major                                         | STEM major                          | 69% (188)         |
|                                               | Non-STEM                            | 29% (80)          |
|                                               | <i>Decline to state</i>             | 1% (4)            |
| Anxiety (GAD-7, mean scores) ^                | Mean (range: 0-3)                   | 0.82              |
|                                               | "+- std"                            | 0.8               |
| Trait anxiety severity (GAD-7, sum scores) ^^ | None or minimal                     | 52% (141)         |
|                                               | Mild                                | 25% (68)          |
|                                               | Moderate                            | 13% (35)          |
|                                               | Severe                              | 10% (26)          |
|                                               | <i>Decline to state</i>             | 1% (2)            |
| Class discipline                              | Biology                             | 68% (186)         |
|                                               | Chemistry                           | 11% (31)          |
|                                               | Physics                             | 14% (39)          |
|                                               | <i>Decline to state</i>             | 6% (16)           |
| Class size                                    | Mean                                | 159.42            |
|                                               | "+- std"                            | 101.69            |

^Mean scores were used in all regression analyses as mean scores are more robust to low levels of missing data (e.g., a participant accidentally skips one item in a 7-item scale) than sum scores (Parent, 2013).

^^ Based on clinical cutoffs of GAD-7 sum scores (Kroenke et al., 2007; Plummer et al., 2016; Spitzer et al., 2006).

**Additional codes for why aspects of groupwork alleviate students' anxiety**

| <b>Themes</b>            | <b>Description</b>                                                       | <b>Example quote</b>                                                                                                                                                             |
|--------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Provides joy             | Student describes feeling happier and joyful.                            | Student 184 (Forming new personal connections): "Making new connections with other students is fun and enjoyable."                                                               |
| Boosts confidence        | Student describes feeling more confident.                                | Student 82 (Receiving validation): "It makes me feel more confident in my abilities in the class."                                                                               |
| Cultivates trust         | Student describes feelings of trust and reliability.                     | Student 229 (Working with student you know): "I much prefer working with others I know as there a sense of camaraderie."                                                         |
| Maintains accountability | Student describes feeling held accountable to stay on track and succeed. | Student 27 (Receiving student academic support): "I worry much less about missing important deadlines or assignments if I have other students that can support me in the class." |

**Additional codes for why aspect of groupwork exacerbate students' anxiety**

| <b>Themes</b>               | <b>Description</b>                                                                                                    | <b>Example quote</b>                                                                                                                                                                                                                                             |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dread exclusion             | Student describes worrying about being rejected, left out and excluded.                                               | Student 34 (Being asked to initiate social interactions): "I don't want to run the risk of being rejected by a group."                                                                                                                                           |
| Engenders hopelessness      | Student describes feelings of hopelessness, lack of belonging and uncertainty about the likelihood of future success. | Student 257 (Realizing you struggle more than others): "It is validating my feelings that I am not worthy to be in college and am going to fail."                                                                                                                |
| Generates external pressure | Student describes feeling an external expectation and pressures from others and the need to meet them.                | Student 268 (Having grade dependent on others' contributions): "What if my part is bad or I didn't do it right then that means [my peers] grade is on me as well and if they are doing really well and their grade drops it's because of me and it is my fault." |

### Additional student recommendations for how instructors can decrease anxiety

| Themes                  | Description                                                                                                                                                                                                              | Percent (n=249) | Example quote                                                                                                                                                                             |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Initiate conversations  | Students recommend breaking the ice, initiating, encouraging and facilitating conversations between group members.                                                                                                       | 8.0% (20)       | Student 18: “Encourage friendly discussions among students.”                                                                                                                              |
| Maintain accountability | Students recommend monitoring groups to maintain accountability, address conflicts or issues and ensure comfort and equal contributions.                                                                                 | 6.8% (17)       | Student 40: “Walk around and make sure that everyone is participating and not letting one person talk and everyone else doesn’t engage.”                                                  |
| Instruct better         | Students recommend practicing, learning, teaching and understanding the content more or better and providing clearer instructions.                                                                                       | 6.4% (16)       | Student 98: “Explaining material better during lectures.”                                                                                                                                 |
| Provide resources       | Students recommend providing support beyond group work, such as holding exam reviews, giving feedback on assignments, more clarity in syllabus, post reminders, offer extra credit, flexibility and extending deadlines. | 5.6% (14)       | Student 21: “Make everything extremely clear in the syllabus.”                                                                                                                            |
| Provide support         | Students recommend providing more support and guidance by checking in, being more involved, interactive and available to answer questions or help.                                                                       | 4.8% (12)       | Student 117: “Be available to answer any questions.”                                                                                                                                      |
| Be personable           | Students recommend being more open, kind, patient, reassuring, forgiving, authentic and not judgmental.                                                                                                                  | 4.8% (12)       | Student 220: “Bring understanding and forgiving. Knowing that I can go to them about problems and that they will be addressed!”                                                           |
| Welcome feedback        | Students recommend giving them opportunities share anonymous feedback and opinions about the course, group, etc. through surveys.                                                                                        | 3.6% (9)        | Student 15: “Initiate a quiz like this to the students beforehand as an anonymous poll to see if group work would be more beneficial to the class than harmful to the student’s anxiety.” |

|                                         |                                                                   |             |                                                            |
|-----------------------------------------|-------------------------------------------------------------------|-------------|------------------------------------------------------------|
| Don't assign or<br>reduce<br>groupwork  | Students recommend not<br>assigning group work or<br>reducing it. | 2.4%<br>(6) | Student 233: "Not assigning group<br>work if possible."    |
| Maintain<br>consistency<br>among groups | Students recommend keeping<br>groups consistent.                  | 2%<br>(5)   | Student 229: "Keep those groups<br>throughout a semester." |

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